Methods of Measuring Trend

Trend is measured using by the following methods:

1. Graphical method
2. Semi averages method
3. Moving averages method
4. Method of least squares

7.3.1 Graphical Method

Under this method the values of a time series are plotted on a graph paper by taking time variable on the X-axis and the values variable on the Y-axis. After this, a smooth curve is drawn with free hand through the plotted points. The trend line drawn above can be extended to forecast the values. The following points must be kept in mind in drawing the freehand smooth curve.

(i) The curve should be smooth

(ii) The number of points above the line or curve should be approximately equal to the points below it

(iii) The sum of the squares of the vertical deviation of the points above the smoothed line is equal to the sum of the squares of the vertical deviation of the points below the line.

Merits

- It is simple method of estimating trend.
- It requires no mathematical calculations.
- This method can be used even if trend is not linear.

Demerits

- It is a subjective method
- The values of trend obtained by different statisticians would be different and hence not reliable.

Example 7.1

Annual power consumption per household in a certain locality was reported below.

Years	1991	1992	1993	1994	1995	1996	1997	1998	1999	2000
Power used (units)	15	20	21	25	28	26	30	32	40	38

Draw a free hand curve for the above data.

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